Relativistic Time Dilation & Coordinate Phase Integration

Complete Spacetime Mathematics of the Time Dilation DAW Engine

Time Dilation DAW (v4.0) Engineering Group

August 3, 2026

Abstract

This document details the mathematical framework governing relativistic coordinate time dilation, Lorentz velocity transformations, Schwarzschild gravitational redshift event horizons, metric time grid quantization, and digital audio phase integration in **Time Dilation DAW (v4.0)**.

1 Introduction to Relativistic Coordinate Time (τ)

In standard Digital Audio Workstations (DAWs), digital time progresses linearly at a constant sample rate f_s:

$$t(n) = \frac{n}{f_s} \tag{1}$$

In **Time Dilation DAW**, coordinate time is modulated by a dynamic relativistic factor $\gamma(t) \in [-16.0, 16.0]$, representing localized gravitational or velocity time dilation:

$$d\tau = \gamma(t)\, dt \tag{2}$$

The accumulated coordinate time $\tau(t)$ after physical elapsed timeline t is obtained by phase integration:

$$\tau(t) = \int_0^t \gamma(t')\, dt' \tag{3}$$

2 Velocity Time Dilation (Special Relativity)

Under Einstein's Special Theory of Relativity, coordinate time dilation for an observer moving at relative velocity $v(t)$ with respect to speed of light c is defined by the Lorentz factor:

$$\gamma(v) = \frac{1}{\sqrt{1 - \frac{v(t)^2}{c^2}}} \tag{4}$$

For $v \to c$, $\gamma(v) \to \infty$, causing asymptotic time dilation. In our engine, γ is clamped to $[-16.0, 16.0]$ for real-time digital stability.

3 Gravitational Time Dilation & Event Horizon Redshift

Near a massive gravitational body of Schwarzschild radius $r_s = \frac{2GM}{c^2}$, time dilation at radius r follows:

$$\gamma(r) = \sqrt{1 - \frac{r_s}{r}} = \text{redshift} \tag{5}$$

As $r \to r_s^+$ (approaching the Event Horizon), redshift $\to 0$, causing gravitational time stasis ($\gamma \to 0.0$). In the `[time.singularity~]` node, non-linear curvature warping is applied:

$$\gamma_{\text{singularity}}(r, M) = \text{clamp}\left(\sqrt{\left|1 - \frac{2GM}{c^2 r}\right|}, 0.0, 16.0\right) \tag{6}$$

4 Retrograde Time Reversal & Metric Grid Quantization

4.1 Retrograde Time Reversal (`[time.retro~]`)

When $\gamma(n) < 0$, the coordinate time rate reverses ($\dot{\tau} = -1.0$), executing authentic *Retrograde Time Reversal*:

$$\tau[n] = \tau[n-1] + \frac{\gamma[n]}{f_s}, \quad \text{where } \gamma[n] \in [-16.0, 0.0) \tag{7}$$

Under retrograde flow, phase increments $\Delta\phi(n) = \frac{2\pi f \gamma(n)}{f_s} < 0$, causing audio buffer read pointers to traverse backwards in real time.

4.2 Metric Time Grid Quantization (`[time.quantize~]`)

To enforce rhythmic micro-step stuttering, coordinate time $\tau(t)$ is discretized against a metric grid division $Q \in \mathbb{R}^+$ (e.g. 1/16-note grid duration $T_Q = \frac{60}{4 \cdot \text{BPM}}$):

$$\tau_{\text{quantized}}(t) = \lfloor \frac{\tau(t)}{T_Q} \rfloor \cdot T_Q + \text{hold}\left(\tau(t) \pmod{T_Q}\right) \tag{8}$$

5 Coordinate Phase Integration in Discrete Sampler Domain

In discrete time at sample rate f_s, the per-sample phase increment $\Delta\phi(n)$ for an oscillator or table reader at frequency f is:

$$\Delta\phi(n) = \frac{2\pi \cdot f \cdot |\gamma(n)|}{f_s} \tag{9}$$

The continuous phase accumulator $\phi[n]$ is updated according to:

$$\phi[n] = (\phi[n-1] + \Delta\phi(n)) \pmod{2\pi} \tag{10}$$

This guarantees phase continuity and exact frequency scaling under arbitrary dynamic time warping $\gamma(t)$.